Aaron McLean

Embedded Software Engineer

aaronmclean26@gmail.com aaronmcleann aaronmcleancs aaronmclean.xyz

Education

Carleton University, Ottawa, Ontario *2021 – 2025*

Bachelor of Computer Science (B.C.S.)

- Recipient of C.J. Mackenzie Scholarship Award
- Coursework and independent project work focused on embedded systems, real-time programming, and low-level computer architecture, forming the technical foundation for industrial automation and control system engineering

Work Experience

Embedded Software Engineer *2025 – Present*

MEDATech Engineering, Collingwood, Ontario

- Architected embedded control system solutions for battery-electric vehicle powertrains (AltDrive) and custom drilling equipment, owning PLC logic (IEC 61131-3 / CoDeSys), safety interlocks, and CAN bus (J1939/CANopen) communication
- Led the development of automated state-machine sequences for specialized equipment (e.g., auto make/break and upper breakout functions), integrating rotary encoders, PID control loops, and feed-forward torque limiting
- Designed and deployed distributed operator interfaces (RRC/HMI), presenting real-time control states, diagnostic fault tables, and automated sequence feedback to field operators
- Engineered full-stack telemetry data loggers using Linux-based Owasys units, implementing UDP packet handling, Wireguard VPN tunnels, and captive portals for high-frequency Modbus and CAN network data aggregation
- Built and maintained automated hardware-in-the-loop (HIL) testing frameworks and dashboards, replacing manual verification with simulated battery load and inverter feedback validation to increase manufacturing throughput

Software Developer *2024 – 2025*

Chimoney, Ottawa, Ontario

- Contributed to a distributed payment API (Node.js) within a remote-first engineering team, implementing subaccount routing and cross-platform integrations, demonstrating foundational backend architecture skills adaptable to scalable industrial platforms

IT Consultant *2023*

Stone Tree Clinic, Collingwood, Ontario

- Designed and deployed a secure network-attached storage (NAS) file-sharing system with role-based access control, applying networking and cybersecurity best practices critical to industrial network infrastructure and remote telemetry security

Technical Projects

Quantum-Resistant Blockchain Consensus Protocol

- Designed a custom consensus protocol replacing proof-of-work with quantum-state fidelity validation, implementing quantum hashing and entanglement verification in Python using IBM Qiskit – demonstrating rigorous low-level systems programming and state-machine/protocol design skills

Embedded Neural Network Deployment

- Developed and optimized a convolutional neural network for diagnostic image classification, adapting it for edge deployment within memory-constrained environments via a mixed-precision inference policy and hardware-accelerated execution

Technical Skills

Controls & Embedded Systems:
IEC 61131-3 (Structured Text), CoDeSys, PLCNext, MATLAB / Simulink, CAN Bus (J1939, CANopen), Modbus, OPC UA, UDP Logging, Hardware-in-the-Loop (HIL) Testing, Vector CANalyzer

Programming Languages:
C, C++, Python, Rust, JavaScript, SQL

Software Development & Tools:
Git, Grafana, AWS, Docker, Wireguard, Linux, JIRA